

Source:

Main.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "bootstrap/dist/css/bootstrap.min.css";
import "bootstrap/dist/js/bootstrap.bundle.min.js";
import "./App.css";
ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
);
```

App.jsx:

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Login from "./pages/Login";
import Register from "./pages/Register";
import Dashboard from "./pages/Dashboard";
import Labs from "./pages/Labs";
import Resources from "./pages/Resources";
import Booking from "./pages/Booking";
import Reports from "./pages/Reports.jsx";
import Profile from "./pages/Profile";
function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Login />} />
        <Route path="/register" element={<Register />} />
        <Route path="/dashboard" element={<Dashboard />} />
        <Route path="/labs" element={<Labs />} />
        <Route path="/resources" element={<Resources />} />
        <Route path="/booking" element={<Booking />} />
        <Route path="/reports" element={<Reports />} />
        <Route path="/profile" element={<Profile />} />
      </Routes>
    </BrowserRouter>
  );
}
```

```
export default App;
```

App.css:

```
body {
  margin: 0;
  background: #f5f7fa;
```

```

        font-family: Arial, Helvetica, sans-serif;
    }

    .page {
        padding: 30px;
    }

    .card-box {
        background: white;
        border-radius: 10px;
        padding: 20px;
        box-shadow: 0px 2px 8px rgba(0,0,0,.1);
    }

    .title {
        color: #0d6efd;
        font-weight: bold;
    }

    .login-container {
        height: 100vh;
        display: flex;
        justify-content: center;
        align-items: center;
    }

    .login-card {
        width: 400px;
        padding: 30px;
        border-radius: 15px;
        background: white;
        box-shadow: 0px 5px 15px rgba(0,0,0,.2);
    }

```

Components:

Navbar.jsx:

```

import { Link } from "react-router-dom";

function Navbar() {
    return (
        <nav className="navbar navbar-expand-lg navbar-dark bg-primary">
            <div className="container-fluid">
                <Link className="navbar-brand fw-bold" to="/dashboard">
                    Lab Resource Utilization Platform
                </Link>

                <button
                    className="navbar-toggler"
                    type="button"

```

```

        data-bs-toggle="collapse"
        data-bs-target="#navbarNav"
      >
        <span className="navbar-toggler-icon"></span>
      </button>

      <div className="collapse navbar-collapse" id="navbarNav">
        <ul className="navbar-nav ms-auto">
          <li className="nav-item">
            <Link className="nav-link"
to="/dashboard">Dashboard</Link>
          </li>

          <li className="nav-item">
            <Link className="nav-link" to="/labs">Labs</Link>
          </li>

          <li className="nav-item">
            <Link className="nav-link"
to="/resources">Resources</Link>
          </li>

          <li className="nav-item">
            <Link className="nav-link" to="/booking">Booking</Link>
          </li>

          <li className="nav-item">
            <Link className="nav-link" to="/reports">Reports</Link>
          </li>

          <li className="nav-item">
            <Link className="nav-link" to="/profile">Profile</Link>
          </li>
        </ul>
      </div>
    </div>
  </nav>
);
}

```

```
export default Navbar;
```

Footer.jsx:

```

function Footer() {
  return (
    <footer className="bg-dark text-white text-center py-3 mt-5">
      © 2026 Lab Resource Utilization Platform | Mepco Schlenk
      Engineering College
    </footer>
  );
}

```

```
}
```

```
export default Footer;
```

Slidebar.jsx:

```
import { Link } from "react-router-dom";
```

```
function Sidebar() {  
  return (  
    <div  
      className="bg-dark text-white p-3"  
      style={{ minHeight: "100vh", width: "220px" }}  
    >  
      <h4 className="text-center mb-4">Menu</h4>  
  
      <div className="d-grid gap-2">  
        <Link className="btn btn-outline-light" to="/dashboard">  
          Dashboard  
        </Link>  
  
        <Link className="btn btn-outline-light" to="/labs">  
          Labs  
        </Link>  
  
        <Link className="btn btn-outline-light" to="/resources">  
          Resources  
        </Link>  
  
        <Link className="btn btn-outline-light" to="/booking">  
          Booking  
        </Link>  
  
        <Link className="btn btn-outline-light" to="/reports">  
          Reports  
        </Link>  
  
        <Link className="btn btn-outline-light" to="/profile">  
          Profile  
        </Link>  
      </div>  
    </div>  
  );  
}
```

```
export default Sidebar;
```

Pages:

Booking.jsx:

```
import { useState } from "react";
```

```

import Navbar from "../components/Navbar";
import Footer from "../components/Footer";
function Booking() {
  const [booking, setBooking] = useState({
    student: "",
    lab: "",
    resource: "",
    date: "",
    time: "",
  });

  const handleChange = (e) => {
    setBooking({ ...booking, [e.target.name]: e.target.value });
  };

  const handleSubmit = (e) => {
    e.preventDefault();

    alert("Booking Request Submitted Successfully!");

    setBooking({
      student: "",
      lab: "",
      resource: "",
      date: "",
      time: "",
    });
  };

  return (
    <>
      <Navbar />

      <div className="container mt-4">
        <div className="card shadow">
          <div className="card-header bg-primary text-white">
            Resource Booking
          </div>

          <div className="card-body">
            <form onSubmit={handleSubmit}>
              <div className="mb-3">
                <label>Student Name</label>
                <input
                  className="form-control"
                  name="student"
                  value={booking.student}
                  onChange={handleChange}
                  required
                />
              </div>
            </form>
          </div>
        </div>
      </div>
    </>
  );
}

```

```
</div>

<div className="mb-3">
  <label>Laboratory</label>
  <select
    className="form-select"
    name="lab"
    value={booking.lab}
    onChange={handleChange}
    required
  >
    <option value="">Select Lab</option>
    <option>Computer Lab</option>
    <option>Electronics Lab</option>
    <option>Communication Lab</option>
  </select>
</div>

<div className="mb-3">
  <label>Resource</label>
  <input
    className="form-control"
    name="resource"
    value={booking.resource}
    onChange={handleChange}
    required
  />
</div>

<div className="mb-3">
  <label>Date</label>
  <input
    type="date"
    className="form-control"
    name="date"
    value={booking.date}
    onChange={handleChange}
    required
  />
</div>

<div className="mb-3">
  <label>Time Slot</label>
  <input
    type="time"
    className="form-control"
    name="time"
    value={booking.time}
    onChange={handleChange}
  >
</div>
```

```

                required
            />
        </div>

        <button className="btn btn-primary">Book
Resource</button>
    </form>
</div>
</div>
</div>
</div>
<Footer />
</>
);
}
export default Booking;

```

Dashboard.jsx:

```

import Navbar from "../components/Navbar";
import Footer from "../components/Footer";
function Dashboard() {
    const stats = [
        {
            title: "Total Labs",
            value: 10,
            color: "primary",
        },
        {
            title: "Resources",
            value: 120,
            color: "success",
        },
        {
            title: "Today's Bookings",
            value: 25,
            color: "warning",
        },
        {
            title: "Available Resources",
            value: 95,
            color: "info",
        },
    ],
    bookings = [
        {
            id: 1,
            student: "Rahul",

```



```

<div className="row">
  <div className="col-md-8">
    <div className="card shadow">
      <div className="card-header bg-primary text-white">
        Recent Bookings
      </div>

      <div className="card-body">
        <table className="table table-bordered table-hover">
          <thead>
            <tr>
              <th>ID</th>
              <th>Student</th>
              <th>Lab</th>
              <th>Resource</th>
              <th>Status</th>
            </tr>
          </thead>

          <tbody>
            {bookings.map((booking) => (
              <tr key={booking.id}>
                <td>{booking.id}</td>

                <td>{booking.student}</td>

                <td>{booking.lab}</td>

                <td>{booking.resource}</td>

                <td>
                  <span
                    className={
                      booking.status === "Approved"
                        ? "badge bg-success"
                        : booking.status === "Pending"
                          ? "badge bg-warning text-dark"
                          : "badge bg-danger"
                    }
                  >
                    {booking.status}
                  </span>
                </td>
              </tr>
            ))}
          </tbody>
        </table>
      </div>
    </div>
  </div>
</div>

```

```

        <div className="col-md-4">
          <div className="card shadow">
            <div className="card-header bg-success text-white">
              System Status
            </div>

            <div className="card-body">
              <p> Server Status : Online</p>

              <p> Database : Connected</p>

              <p> Active Users : 42</p>

              <p> Booking Requests : 8</p>

              <p> Available Labs : 10</p>
            </div>
          </div>
        </div>
      </div>
    </div>
    <Footer />
  </>
);
}
export default Dashboard;

```

Labs.jsx:

```

import Navbar from "../components/Navbar";
import Footer from "../components/Footer";
function Labs() {
  const labs = [
    {
      id: 1,
      name: "Computer Lab",
      location: "Block A",
      capacity: 60,
      status: "Available",
    },
    {
      id: 2,
      name: "Electronics Lab",
      location: "Block B",
      capacity: 40,
      status: "Occupied",
    },
  ];
}

```

```

        id: 3,
        name: "Communication Lab",
        location: "Block C",
        capacity: 35,
        status: "Available",
    },
    {
        id: 4,
        name: "Microprocessor Lab",
        location: "Block D",
        capacity: 30,
        status: "Occupied",
    },
    {
        id: 5,
        name: "Digital Systems Lab",
        location: "Block E",
        capacity: 25,
        status: "Available",
    },
    {
        id: 6,
        name: "Control Systems Lab",
        location: "Block F",
        capacity: 20,
        status: "Occupied",
    },
    {
        id: 7,
        name: "Power Electronics Lab",
        location: "Block G",
        capacity: 15,
        status: "Available",
    },
];

return (
    <>
        <Navbar />

        <div className="container mt-4">
            <h2 className="text-primary mb-4">Laboratories</h2>

            <table className="table table-bordered table-hover shadow">
                <thead className="table-primary">
                    <tr>
                        <th>ID</th>
                        <th>Lab Name</th>
                        <th>Location</th>

```

```

        <th>Capacity</th>
        <th>Status</th>
      </tr>
    </thead>

    <tbody>
      {labs.map((lab) => (
        <tr key={lab.id}>
          <td>{lab.id}</td>
          <td>{lab.name}</td>
          <td>{lab.location}</td>
          <td>{lab.capacity}</td>
          <td>
            <span
              className={`badge ${
                lab.status === "Available" ? "bg-success" : "bg-
danger"
              }`}
            >
              {lab.status}
            </span>
          </td>
        </tr>
      )
    )}
    </tbody>
  </table>
</div>

<Footer />
</>
);
}
export default Labs;

```

Login.jsx

```

import { useState } from "react";
import { Link, useNavigate } from "react-router-dom";

function Login() {

  const navigate = useNavigate();

  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");

  const login = (e) => {

    e.preventDefault();

    const users =

```

```

    JSON.parse(localStorage.getItem("users")) || [];

const validUser = users.find(
  (user) =>
    user.email === email &&
    user.password === password
);

if (validUser) {
  localStorage.setItem(
    "loggedInUser",
    JSON.stringify(validUser)
  );

  alert("Login Successful");

  navigate("/dashboard");
} else {
  alert("Invalid Email or Password");
}
};

return (
  <div className="login-container">
    <div className="login-card">
      <h2 className="text-center text-primary mb-4">
        Login
      </h2>
      <form onSubmit={login}>
        <div className="mb-3">
          <label>Email</label>
          <input
            className="form-control"
            type="email"
            placeholder="Enter Email"
            value={email}
            onChange={(e)=>setEmail(e.target.value)}
            required
          />

```

```

    </div>

    <div className="mb-3">

      <label>Password</label>

      <input
        className="form-control"
        type="password"
        placeholder="Enter Password"
        value={password}
        onChange={(e)=>setPassword(e.target.value)}
        required
      />

    </div>

    <button
      type="submit"
      className="btn btn-primary w-100"
    >
      Login
    </button>

  </form>

  <div className="text-center mt-3">

    Don't have an account?

    <br />

    <Link to="/register">
      Register Here
    </Link>

  </div>

</div>

</div>

);
}

```

```
export default Login;
```

Profile.jsx:

```

import { useState } from "react";
import Navbar from "../components/Navbar";
import Footer from "../components/Footer";

```

```

function Profile() {
  const [user, setUser] = useState({
    name: "Senthil Murugan",
    email: "senthil@gmail.com",
    department: "ECE",
    phone: "9876543210",
  });

  const handleChange = (e) => {
    setUser({
      ...user,
      [e.target.name]: e.target.value,
    });
  };

  const saveProfile = (e) => {
    e.preventDefault();
    alert("Profile Updated Successfully");
  };

  return (
    <>
      <Navbar />

      <div className="container mt-4">
        <div className="card shadow">
          <div className="card-header bg-primary text-white">User
Profile</div>

          <div className="card-body">
            <form onSubmit={saveProfile}>
              <div className="mb-3">
                <label>Name</label>
                <input
                  className="form-control"
                  name="name"
                  value={user.name}
                  onChange={handleChange}
                />
              </div>

              <div className="mb-3">
                <label>Email</label>
                <input
                  className="form-control"
                  name="email"
                  value={user.email}
                  onChange={handleChange}
                />
              </div>
            </form>
          </div>
        </div>
      </div>
    </>
  );
}

```

```

        </div>

        <div className="mb-3">
          <label>Department</label>
          <input
            className="form-control"
            name="department"
            value={user.department}
            onChange={handleChange}
          />
        </div>

        <div className="mb-3">
          <label>Phone</label>
          <input
            className="form-control"
            name="phone"
            value={user.phone}
            onChange={handleChange}
          />
        </div>

        <button className="btn btn-success">Update
Profile</button>
      </form>
    </div>
  </div>
  </div>

  <Footer />
</>
);
}
export default Profile;

```

Register.jsx:

```

import { useState } from "react";
import { Link, useNavigate } from "react-router-dom";

function Register() {
  const navigate = useNavigate();

  const [form, setForm] = useState({
    name: "",
    email: "",
    phone: "",
    department: "",
    password: "",
    confirmPassword: "",
  });

```

```

});

const changeHandler = (e) => {
  setForm({
    ...form,
    [e.target.name]: e.target.value,
  });
};

const register = (e) => {
  e.preventDefault();

  if (
    !form.name ||
    !form.email ||
    !form.phone ||
    !form.department ||
    !form.password ||
    !form.confirmPassword
  ) {
    alert("Please fill all fields");
    return;
  }

  if (form.password !== form.confirmPassword) {
    alert("Passwords do not match");
    return;
  }

  // Get existing users
  const users = JSON.parse(localStorage.getItem("users")) || [];

  // Check duplicate email
  const existingUser = users.find(
    (user) => user.email === form.email
  );

  if (existingUser) {
    alert("Email already registered");
    return;
  }

  // Create new user
  const newUser = {
    name: form.name,
    email: form.email,
    phone: form.phone,
    department: form.department,
    password: form.password,
  };
};

```

```

    users.push(newUser);

    localStorage.setItem("users", JSON.stringify(users));

    alert("Registration Successful");

    navigate("/");
  };

  return (
    <div className="container mt-5">
      <div className="row justify-content-center">

        <div className="col-md-6">

          <div className="card shadow">

            <div className="card-body">

              <h2 className="text-center text-primary mb-4">
                Register
              </h2>

              <form onSubmit={register}>

                <input
                  className="form-control mb-3"
                  placeholder="Name"
                  name="name"
                  value={form.name}
                  onChange={changeHandler}
                />

                <input
                  className="form-control mb-3"
                  placeholder="Email"
                  type="email"
                  name="email"
                  value={form.email}
                  onChange={changeHandler}
                />

                <input
                  className="form-control mb-3"
                  placeholder="Phone Number"
                  name="phone"
                  value={form.phone}
                  onChange={changeHandler}
                />
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>
  );
}

```

```

        <input
            className="form-control mb-3"
            placeholder="Department"
            name="department"
            value={form.department}
            onChange={changeHandler}
        />

        <input
            className="form-control mb-3"
            placeholder="Password"
            type="password"
            name="password"
            value={form.password}
            onChange={changeHandler}
        />

        <input
            className="form-control mb-3"
            placeholder="Confirm Password"
            type="password"
            name="confirmPassword"
            value={form.confirmPassword}
            onChange={changeHandler}
        />

        <button
            type="submit"
            className="btn btn-success w-100"
        >
            Register
        </button>

    </form>

    <div className="text-center mt-3">
        Already have an account?
        <br />
        <Link to="/">Login</Link>
    </div>

</div>

</div>

</div>

</div>
);

```

```
}
```

```
export default Register;
```

Reports.jsx:

```
import Navbar from "../components/Navbar";
```

```
import Footer from "../components/Footer";
```

```
function Reports() {
```

```
  const reports = [
```

```
    {
```

```
      id: 1,
```

```
      name: "Laboratory Usage Report",
```

```
      date: "28-07-2026",
```

```
      status: "Generated",
```

```
    },
```

```
    {
```

```
      id: 2,
```

```
      name: "Resource Availability Report",
```

```
      date: "27-07-2026",
```

```
      status: "Generated",
```

```
    },
```

```
    {
```

```
      id: 3,
```

```
      name: "Booking History Report",
```

```
      date: "26-07-2026",
```

```
      status: "Generated",
```

```
    },
```

```
  ];
```

```
  return (
```

```
    <>
```

```
      <Navbar />
```

```
      <div className="container mt-4">
```

```
        <h2 className="text-primary mb-4">Reports</h2>
```

```
        <table className="table table-bordered table-hover shadow">
```

```
          <thead className="table-dark">
```

```
            <tr>
```

```
              <th>ID</th>
```

```
              <th>Report Name</th>
```

```
              <th>Date</th>
```

```
              <th>Status</th>
```

```
              <th>Action</th>
```

```
            </tr>
```

```
          </thead>
```

```
          <tbody>
```

```
            {reports.map((report) => (
```

```
              <tr key={report.id}>
```

```

        <td>{report.id}</td>

        <td>{report.name}</td>

        <td>{report.date}</td>

        <td>
            <span className="badge bg-
success">{report.status}</span>
        </td>

        <td>
            <button className="btn btn-primary btn-
sm">Download</button>
        </td>
    </tr>
    )})
</tbody>
</table>
</div>

<Footer />
</>
);
}
export default Reports;

```

Resources.jsx:

```

import Navbar from "../components/Navbar";
import Footer from "../components/Footer";
function Resources() {
    const resources = [
        {
            id: 1,
            name: "Desktop PC",
            lab: "Computer Lab",
            quantity: 50,
            status: "Available",
        },
        {
            id: 2,
            name: "Oscilloscope",
            lab: "Electronics Lab",
            quantity: 12,
            status: "Available",
        },
        {
            id: 3,
            name: "Signal Generator",

```



```

        <td>{item.name}</td>
        <td>{item.lab}</td>
        <td>{item.quantity}</td>
        <td>
          <span
            className={`badge ${
              item.status === "Available"
                ? "bg-success"
                : "bg-warning text-dark"
            }`}
          >
            {item.status}
          </span>
        </td>
      </tr>
    ))}
  </tbody>
</table>
</div>

<Footer />
</>
);
}
export default Resources;

```

Services:

Api.js:

```

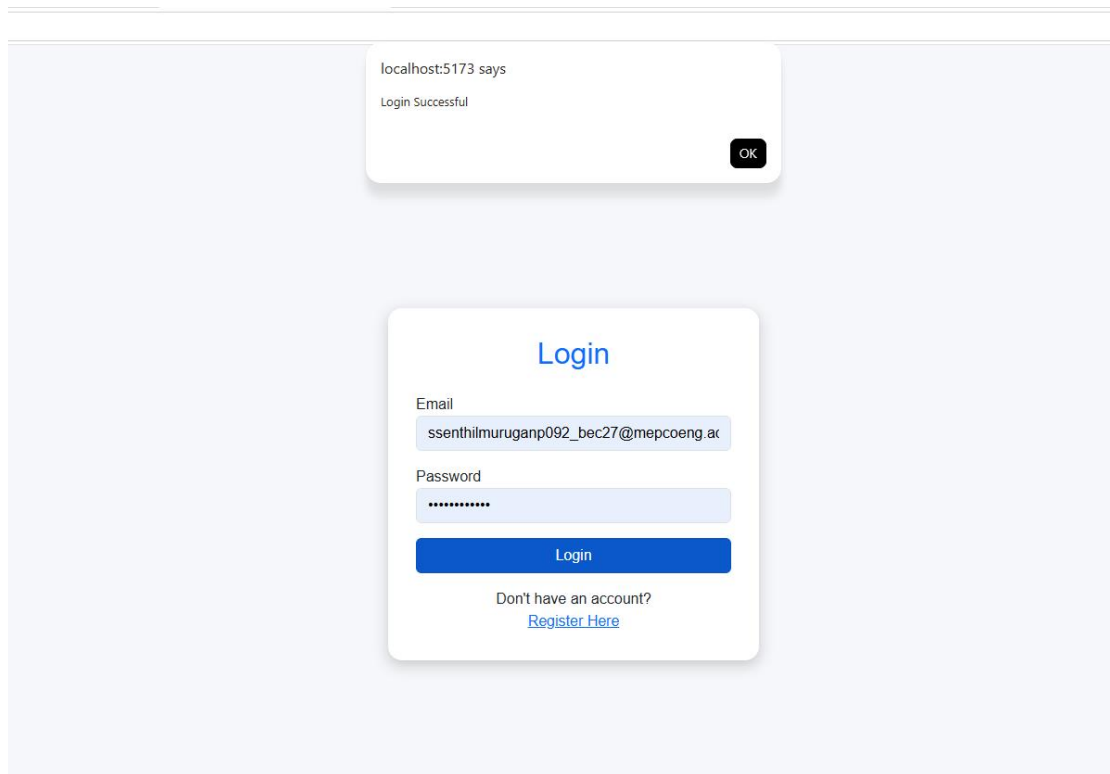
import axios from "axios";

const API = axios.create({
  baseURL: "http://localhost:8080/api"
});

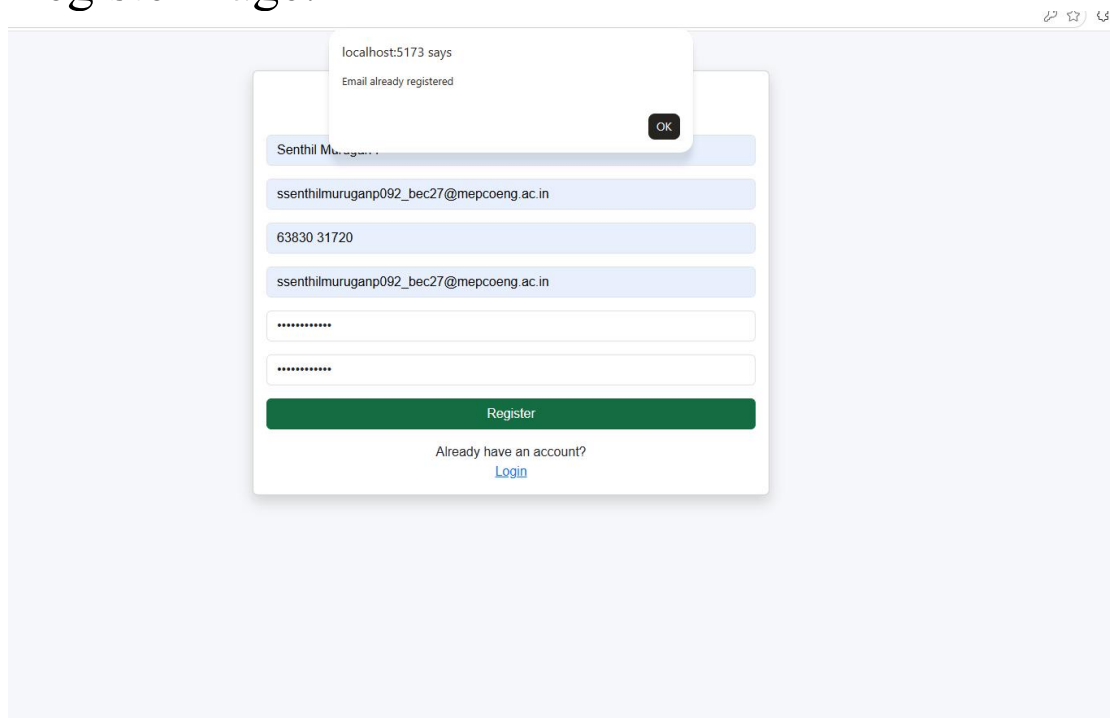
export default API;

```

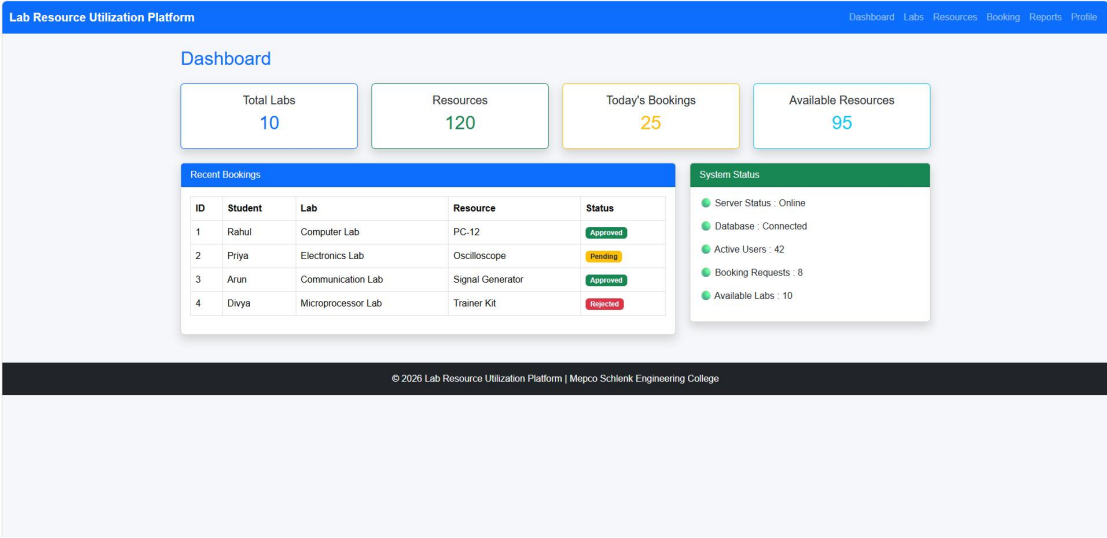
Login page:



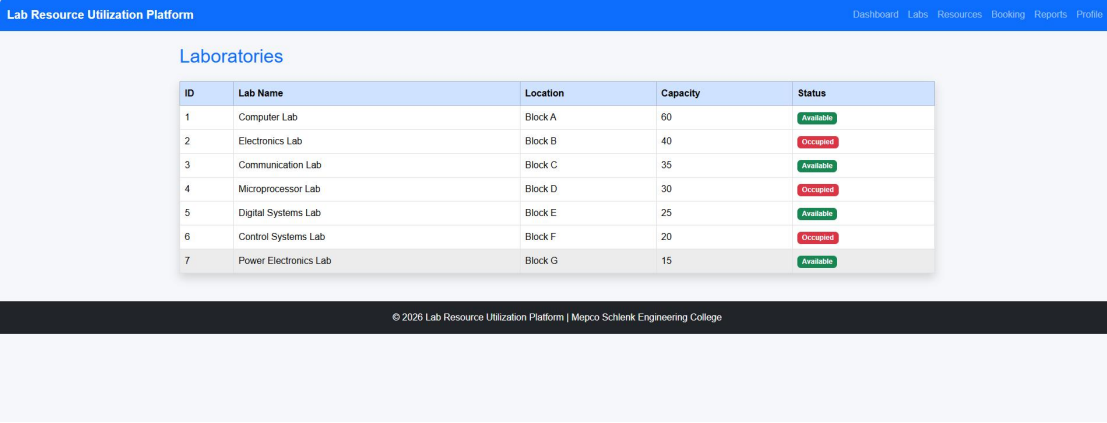
Register Page:



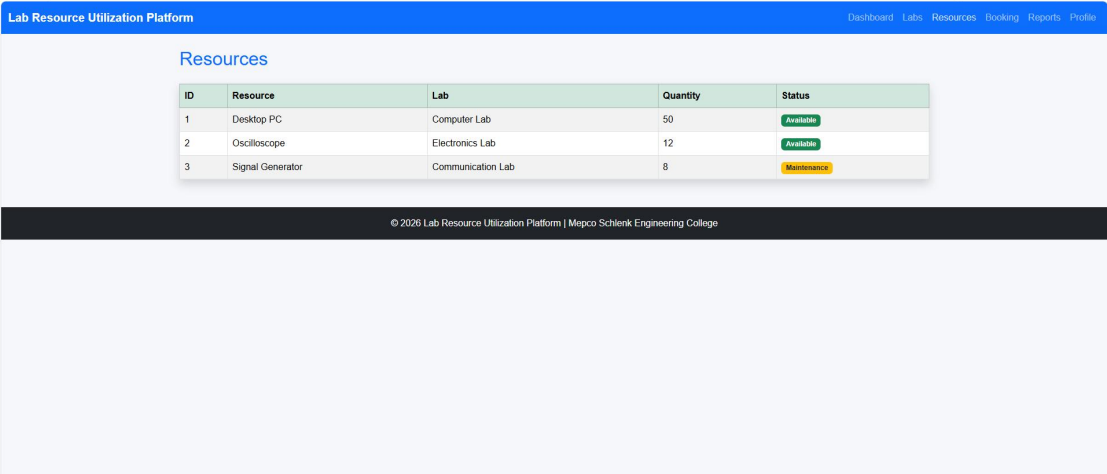
Dashboard:



Labs Page:



Resources:



Booking Page:

Lab Resource Utilization Platform

Dashboard Labs Resources Booking Reports Profile

Resource Booking

Student Name

Laboratory

Select Lab

Resource

Date

mm/dd/yyyy

Time Slot

--:-- --

Book Resource

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Report Page:

Lab Resource Utilization Platform

Dashboard Labs Resources Booking Reports Profile

Reports

ID	Report Name	Date	Status	Action
1	Laboratory Usage Report	26-07-2026	Generated	Download
2	Resource Availability Report	27-07-2026	Generated	Download
3	Booking History Report	26-07-2026	Generated	Download

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Profile Page:

Lab Resource Utilization Platform

Dashboard Labs Resources Booking Reports Profile

User Profile

Name

Senthil Murugan

Email

senthil@gmail.com

Department

ECE

Phone

9876543210

Update Profile

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Database:

File Edit View Query Database Server Tools Scripting Help

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context help Snippets

Output

Action Output

Time Action Message Duration / Fetch

Object Info Session

SCHEMAS

Filter objects

lab_resource_db

Tables

bookings

Columns

Indices

ForeignKeys

Triggers

laboratories

Columns

Indices

ForeignKeys

Triggers

maintenance

Columns

Indices

ForeignKeys

Triggers

resources

Columns

Indices

ForeignKeys

Triggers

users

Columns

Indices

ForeignKeys

Triggers

Administration Schemas

Information

Table: bookings

Columns:

booking_id INT AI PK

user_id INT

resource_id INT

booking_date DATE

start_time TIME

end_time TIME

quantity INT

status ENUM('PENDING')

```
25 category VARCHAR(50),
26 total_quantity INT,
27 available_quantity INT,
28 lab_id INT,
29 FOREIGN KEY (lab_id) REFERENCES laboratories(lab_id)
30 );
31
32
33 CREATE TABLE bookings (
34 booking_id INT AUTO_INCREMENT PRIMARY KEY,
35 user_id INT,
36 resource_id INT,
37 booking_date DATE,
38 start_time TIME,
39 end_time TIME,
40 quantity INT,
41 status ENUM('PENDING','APPROVED','REJECTED','CANCELLED') DEFAULT 'PENDING',
42 FOREIGN KEY (user_id) REFERENCES users(user_id),
43 FOREIGN KEY (resource_id) REFERENCES resources(resource_id)
44 );
45
46 CREATE TABLE maintenance (
47 maintenance_id INT AUTO_INCREMENT PRIMARY KEY,
48 resource_id INT,
49 issue_description TEXT,
50 reported_by INT,
51 reported_date DATE,
52 status ENUM('OPEN','IN_PROGRESS','RESOLVED') DEFAULT 'OPEN',
53 FOREIGN KEY (resource_id) REFERENCES resources(resource_id),
54 FOREIGN KEY (reported_by) REFERENCES users(user_id)
55 );
```